

JavaScript mid-term exam RETAKE (2020-21-1)

Due No Due Date **Points** None

Important information

- You have **120 minutes** to complete the exam. We provide you **additional 30 minutes** for filling in the `README.md` (see template below), reading the tasks, downloading the materials, compressing and uploading your solution.
- Solutions must be uploaded to ELTE Canvas. **The system closes at 18:30 (Hungarian time) and there will be no opportunity to hand in your solution after the aforementioned time.**
- You may use **any kind of material** (including documentation, lectures, class exercises, cheat sheets, etc) during the exam, **except human help** (synchronous, asynchronous, chat, forums, etc)!
- The tasks are independent and can be solved **in any order**.
- To begin your work you should first [download the starting package we have prepared](#). Each task has its own directory in the package. All tasks are somewhat prepared. Generally, you only have to edit the provided `.js` files, but feel free to divide up your solution into multiple `.js` files, or even edit the HTML if needed.
- The starting package contains a template called `README.md` where you should fill in your name and Neptun code (the `<>` characters must be removed). **Submissions without a properly filled `README.md` file will not be graded!**

```
<Name>
<Neptun ID>
This solution was submitted and prepared by the student stated above for the assignment of the Web programming course.
I declare that this solution is my own work.
I have not copied or used third party solutions.
I have not passed my solution to my classmates, neither made it public.
Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C. § ) states
that as long as a student presents another student's work - or at least the significant part of it -
as his/her own performance, it will count as a disciplinary fault. The most serious consequence of a
disciplinary fault can be dismissal of the student from the University.
```

- Each directory contains a file named `TASKS.md`. Replace the space in `[ ]` with an `x` where you have (even partially) succeeded to complete the task. This will make us easier to find your solution when we grade your exam.

Task 1: Array (8 points)

In this task, you have to solve problems on an array of numbers. We recommend using array functions for easy one-liner solutions.

- a. (1 point) The input element with the id `numbers` can be used to create the array. Numbers should be

separated by commas. Read in the value of the field while typing (`input` event) and split it into an array at the commas. Log the resulting array on the console!

- b. (1 point) Right now, all array elements are in string format. Transform all elements into numbers and log the resulting array on the console! Use this array of numbers for subtasks c-g. If you couldn't do the transformation, use an array of numbers pasted into the source code.
- c. (1 point) Count the elements divisible by three and display the result in the element with id `task1`!
- d. (1 point) Output a negative number from the array into the element with id `task2`! If no negative number is present, display the message *"No negative number"*.
- e. (1 point) If all numbers in the array are odd, display the text *"All numbers are odd"* in the element with id `task3`. Otherwise, display the text *"There's at least one even number"*.
- f. (1 point) Display the highest number from the array in the element with id `task4`!
- g. (1 point) Output all unique numbers into the element with id `task5`, for example: `[1,2,2,3,1] --> 1,2,3`
- h. (0.5 points) If an invalid value is entered into the input field, add the class `error` to it! Hint: check the function `isNaN` on all elements!
 - Correct inputs: `1,2,3`, `1, 2, 3`, `1,,3`, `1,,3,,`
 - Incorrect inputs: `apple`, `1,a,3`, `1,a,3a`
- i. (0.5 points): If the input has two commas following each other (e.g `1,,2`), you get empty elements in your array. Filter these out before solving the problems!

5,3,6,6,3,4,2

Task 1. Number of numbers divisible by three: 4

Task 2. A negative number: No negative number

Task 3. Are all numbers odd?: Evens also

Task 4. Largest value: 6

Task 5. Unique numbers: 5,3,6,4,2

Task 2: Images (8 points)

A number of image URLs are provided. We want to view them and select some of them. The URLs are in a `textarea` field, each URL on a separate line, separated by newline characters (`\n`). When clicking a button, we want to display the URLs in a multiple selection field (`select`). Then, we can select one or more (with the Ctrl key) images. The selected images appear in a smaller size below this input. When hovering over a small image, the original appears below. The border-width of this big image is controllable with a range slider.

- a. (2 points) When the `button-to-select` button is clicked, copy the lines from the `textarea` to the `select` element!
- b. (2 points) When selecting an image's URL in the selection box, it appears in the `div` with id `images`.
- c. (1 point) When selecting multiple images' URL's (with the Ctrl key), all of them appear in the `div` with

id `images`.

- d. (1 point) Hovering over a small image displays it in the element with id `big`.
- e. (2 points) Moving the range slider `border` changes the border-width of the big image.

Images

```
https://images.unsplash.com/photo-1531512073830-ba890ca4eba2?h=200&w=300
https://images.unsplash.com/photo-1543753585-4430dfdf72e3?h=200&w=300
https://images.unsplash.com/photo-1465935343323-d742334bcba?h=200&w=300
https://images.unsplash.com/photo-1501884428012-aa321a256730?h=200&w=300
https://images.unsplash.com/photo-1501854140801-50d01698950b?h=200&w=300
https://images.unsplash.com/photo-1517782902678-70c53f6dd2b9?h=200&w=300
```

Transform to select

```
https://images.unsplash.com/photo-1531512073830-ba890ca4eba2?h=200&w=300
https://images.unsplash.com/photo-1543753585-4430dfdf72e3?h=200&w=300
https://images.unsplash.com/photo-1465935343323-d742334bcba?h=200&w=300
https://images.unsplash.com/photo-1501884428012-aa321a256730?h=200&w=300
https://images.unsplash.com/photo-1501854140801-50d01698950b?h=200&w=300
https://images.unsplash.com/photo-1517782902678-70c53f6dd2b9?h=200&w=300
```

Border width:

Use Ctrl to select multiple items



Task 3. Tiaravirus (8 points)

A new pandemic, called Tiaravirus is infecting people. Your task is to create the weekly statistics for new infections using canvas.

- a. (1 point) Generate an array of 7 random numbers between 50 and 350. This is the daily new infections for the last week. Display this table on the console!
- b. (2 points) Display the values of this array on a diagram according to the example GIF! It will fit the background if the starting x coordinate of the first segment is 100, then 200 for the second segment, then 300, etc. The y coordinate is based on the value in the array, but mind the canvas' coordinate system

and the values appearing on the background!

- c. (1 point) The segments should be blue and 3px wide.
- d. (1 point) After the last blue segment, attach another one, but color it red! This is displaying today's new infections. The starting value for today is the same as the last day in your array.
- e. (1 point) Clicking the and buttons increases or decreases today's infections by 10. Update the diagram after each button click!
- f. (1 point) Clicking the button takes us to the next day, so the whole diagram moves to the left (leaving out the oldest day).
- g. (1 point) Clicking the button automatically advances to the next day every 0.5 seconds. When in this mode, the new infections should always be half of yesterday's numbers (simulating an exponential falloff).

Tiaravirus



Task 4. Parity (8 points)

In telecommunication networks, the easiest way of checking for errors is introducing a parity bit: it tells us if the sent data had an even or odd number of 1 bits. In this task, you have to generate parity bits for rows in a table where each cell can have two possible values: black or white.

- a. (2 points) Clicking the button should generate a table based on the number input: for a value of N, generate a table of $(N+1) \times (N+1)$ size. Display this table and unhide the button!
- b. (1 point) The upper left cell should remain empty. In the first row, starting from the second column, display the A, B, C... labels. (Hint: one possible solution is using `String.fromCharCode(65)`)
- c. (1 point) In the first column, starting from the second row, display the numbers 1,2,3,...!
- d. (1 point) All other cells should be empty and styled with the `.border` class.
- e. (1 point) When clicking on a cell with the class `.border` on it, change it's bacground color from white

to black or from black to white!

- f. (1 point) Clicking the **Extend table** button should extend our table by one column. The label above this column should be correct! (So after generating a 5×5 table, the first extra column will have the label F.) If needed, you can regenerate the whole table, but keep the colors!
- g. (1 point) When adding the new cells, set their color so after extending, each row still contains an even number of black cells!

Parity check

	A	B	C	D
1				
2				
3				
4				

Extend table

JS retake

Criteria	Ratings		Pts
Task 1 Subtask a) a. (1 point) The input element with the id numbers can be used to create the array. Numbers should be separated by commas. Read in the value of the field while typing (input event) and split it into an array at the commas. Log the resulting array on the console!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 1 Subtask b) b. (1 point) Right now, all array elements are in string format. Transform all elements into numbers and log the resulting array on the console! Use this array of numbers for subtasks c-g. If you couldn't do the transformation, use an array of numbers pasted into the source code.	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 1 Subtask c) c. (1 point) Count the elements divisible by three and display the result in the element with id task1!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 1 Subtask d) d. (1 point) Output a negative number from the array into the element with id task2! If no negative number is present, display the message "No negative number".	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts

Task 1 Subtask e) e. (1 point) If all numbers in the array are odd, display the text "All numbers are odd" in the element with id task3. Otherwise, display the text "There's at least one even number".	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 1 Subtask f) f. (1 point) Display the highest number from the array in the element with id task4!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 1 Subtask g) g. (1 point) Output all unique numbers into the element with id task5, for example: [1,2,2,3,1] --> 1,2,3	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 1 Subtask h) h. (0.5 points) If an invalid value is entered into the input field, add the class error to it! Hint: check the function isNaN on all elements! Correct inputs: 1,2,3, 1, 2, 3, 1,,3, 1,,3,, Incorrect inputs: apple, 1,a,3, 1,a,3a	0.5 pts Teljes megoldás	0.0 pts Nem értékelhető	0.5 pts
Task 1 Subtask i) i. (0.5 points): If the input has two commas following each other (e.g 1,,2), you get empty elements in your array. Filter these out before solving the problems!	0.5 pts Teljes megoldás	0.0 pts Nem értékelhető	0.5 pts
Task 2 Subtask a) a. (2 points) When the button-to-select button is clicked, copy the lines from the textarea to the selectelement!	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
Task 2 Subtask b) b. (2 points) When selecting an image's URL in the selection box, it appears in the div with id images.	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
Task 2 Subtask c) c. (1 point) When selecting multiple images' URL's (with the Ctrl key), all of them appear in the div with id images.	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 2 Subtask d) d. (1 point) Hovering over a small image displays it in the element with id big.	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts

Task 2 Subtask e) e. (2 points) Moving the range slider border changes the border-width of the big image.	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
Task 3 Subtask a) a. (1 point) Generate an array of 7 random numbers between 50 and 350. This is the daily new infections for the last week. Display this table on the console!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 3 Subtask b) b. (2 points) Display the values of this array on a diagram according to the example GIF! It will fit the background if the starting x coordinate of the first segment is 100, then 200 for the second segment, then 300, etc. The y coordinate is based on the value in the array, but mind the canvas' coordinate system and the values appearing on the background!	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts
Task 3 Subtask c) c. (1 point) The segments should be blue and 3px wide.	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 3 Subtask d) d. (1 point) After the last blue segment, attach another one, but color it red! This is displaying today's new infections. The starting value for today is the same as the last day in your array.	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 3 Subtask e) e. (1 point) Clicking the + and - buttons increases or decreases today's infections by 10. Update the diagram after each button click!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 3 Subtask f) f. (1 point) Clicking the Next button takes us to the next day, so the whole diagram moves to the left (leaving out the oldest day).	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 3 Subtask g) g. (1 point) Clicking the Simulate button automatically advances to the next day every 0.5 seconds. When in this mode, the new infections should always be half of yesterday's numbers (simulating an exponential falloff).	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 4 Subtask a) a. (2 points) Clicking the start button should generate a table based on the number input: for a value of N, generate a table of $(N+1) \times (N+1)$ size. Display this table	2.0 pts Teljes megoldás	0.0 pts Nem értékelhető	2.0 pts

and unhide the Extend button!			
Task 4 Subtask b) b. (1 point) The upper left cell should remain empty. In the first row, starting from the second column, display the A, B, C... labels. (Hint: one possible solution is using <code>String.fromCharCode(65)</code>)	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 4 Subtask c) c. (1 point) In the first column, starting from the second row, display the numbers 1,2,3,...!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 4 Subtask d) d. (1 point) All other cells should be empty and styled with the <code>.border</code> class.	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 4 Subtask e) e. (1 point) When clicking on a cell with the class <code>.border</code> on it, change it's background color from white to black or from black to white!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 4 Subtask f) f. (1 point) Clicking the Extend table button should extend our table by one column. The label above this column should be correct! (So after generating a 5×5 table, the first extra column will have the label F.) If needed, you can regenerate the whole table, but keep the colors!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Task 4 Subtask g) g. (1 point) When adding the new cells, set their color so after extending, each row still contains an even number of black cells!	1.0 pts Teljes megoldás	0.0 pts Nem értékelhető	1.0 pts
Total Points: 32.0			